PROJECT REPORT ON

INSIDER THREAT PREDICTOR SYSTEM

Submitted by

Chittawar Anushri - AD23 - ST#IS#8429

Under the Supervision of

KRISHNA
Security Analyst

Registered And Head Office
D.NO: 11-9-18, 1st Floor,
Majjivari Street, Kothapeta, Vijayawada - 520001.

+91 9550055338 / +91 7901336873

contact@suprajatechnologies.com

COMPANY INTRODUCTION:

Supraja Technologies is a leading Knowledge and Technical Solutions Provider and pioneer leader in IT industry, is operating based out of Vijayawada.

R&D at Supraja

With a 24X7 work in Research & Development, experts at Supraja Technologies work under our :

- **Supraja Technologies Cyber Security Cell**

About Supraja Technologies:

Supraja Technologies (a unit of CHSMRLSS Technologies Pvt. Ltd.) with its foundation pillars as Innovation, Information and Intelligence is exploring indefinitely as a **Technology Service Provider (Corporate Consulting)** and as a **Training Organization (Ed-Tech)** as well.

You may visit us at :
www.suprajatechnologies.com

The multi domains of trainings which Supraja Technologies operate include the following :

- **Workshops & Hackathons**
  - Engineering Colleges
  - Corporate Companies (Startups & MNC's)
  - Government Organizations

- **Classroom Trainings Cum Certification Courses**

○ Summer Training (30-45 Days) ○ Winter Training (10 - 15 Days) ○ Weekend Training (2 Days) ○ 1 Month / 3 Months / 6 Months Courses

- **On-site Trainings**

○ Value Added Courses / Two Credit Courses for Colleges
○ Faculty Development Programs
○ College Summer Training (15 Days, 30 Days, 45 D ays & 60 Days)
○ Govt Agencies, Police Academies, Corporate s etc

- **Cloud Campus**

○ (Distance Learning Program) *Coming Soon

- **Internships**

○ Internship for Engineering Students (30 Days, 45 Days & 60 Days)
○ Internship for Graduates (6 Months)

- **Lab Setup**

○ Cyber Lab

- **CoE**

○ Cyber Security Centre of Excellence

- **Supraja Technologies Security Assessment Product**

○ (Our SaS Product is currently under development) *Coming Soon

Why Supraja Technologies:

Be it Training or a workshop, the course content is always from R&D Cell of Supraja.

- A proven track record of delivering quality services.
- **68,500+** Students trained by our trainers till date.
- Training Partners of recognized institutions.
- Trainers with excellent research and teaching pedagogy illustrate their findings through corporate standard **practical demonstrations** during their sessions.
- Easy to learn and **hands-on sessions** are given, with additional benefits of Study Material, Tool kit and immediate query handling.
- Self-Prepared **Cyber Security Cell.**
- Supraja Technologies has the best, experienced and highly **skilled bunch of R&D Engineers, Security Analysts, Security Consultants & Trainers.**
- We provide training in Innovating and Trending Technologies to Govt. Officials, Corporate Houses and Colleges.

Something we are proud of :

1. Supraja Technologies CEO Mr.Santosh Chaluvadi is an Alumni of Potti Sriramulu Chalavadi Mallikharjuna Rao College of Engineering and Technology, Vijayawada.

   With our CEO this college conducted/organised a 50 hours Nonstop Marathon Training Workshop on Ethical Hacking & Cyber Security for which this respective college and our CEO both holds their name in **“LIMCA BOOK OF RECORDS 2017”**

2. We are very happy to inform you all that our company, Supraja Technologies has been shortlisted for **"Top 50 Tech Companies" award 2019**, conferred at InterCon - Dubai, UAE.

Supraja Technologies is one out of thousands companies that were initially screened by InterCon team of 45+ research analysts over a period of three months and the final shortlist includes 150+ firms and we are very proud to inform you all that our company Supraja Technologies also happens to be a part of the same.

Life changing solution/service :

After working on R&D for around 2 years, finally in the mid 2019 we have successfully developed a service/solution of various techniques and strategies for the Film Industry through which he can kill piracy of any film in online up to 35% right now. This betaservice is being appreciated & adopted by various Tollywood Film Industry Producers & Hero's to safeguard their film from piracy in online and to gain more profits.

By the end of 2033 our vision is to rollout a complete full packed service/solution where we can kill piracy entirely 100% everywhere in online for sure.

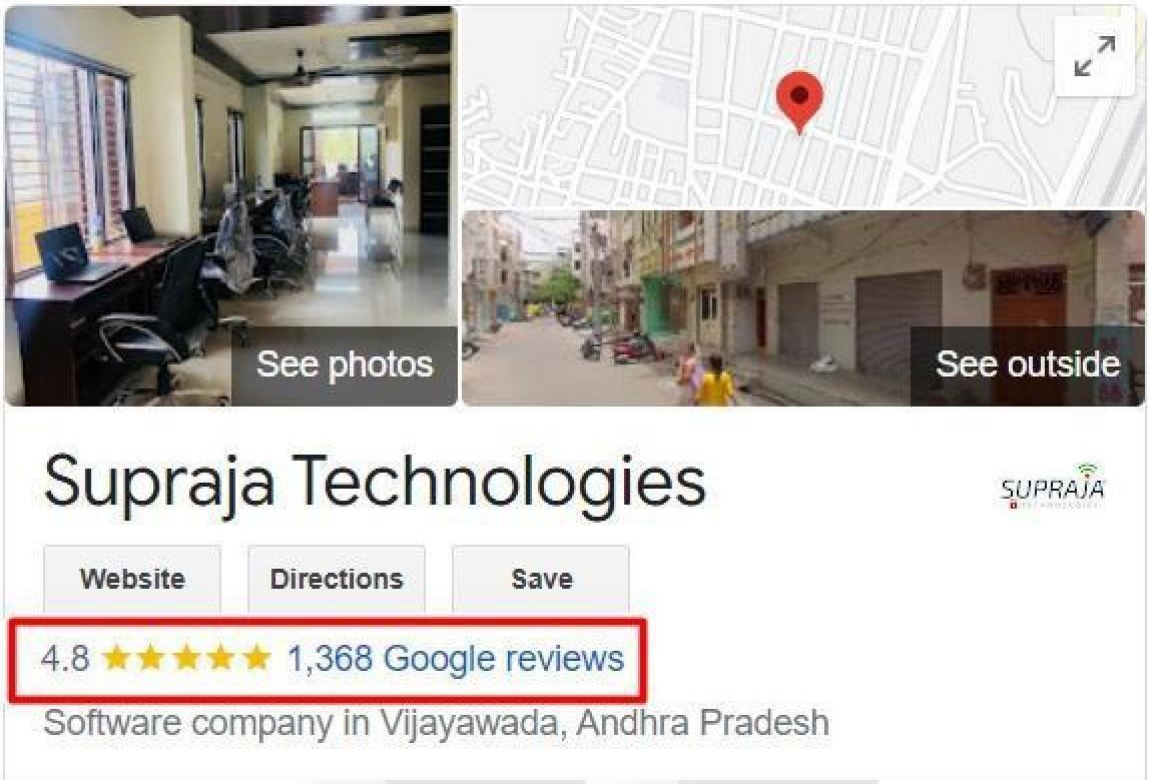
Appreciation :

Received a great appreciation from our 1st Tollywood Film Industry client Mr.Saptagiri for providing our Anti-Piracy betaservice for his film VAJRA KAVACHADARA GOVINDA

Achievements:

- On 17th August 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security** (CoE) at Ramco Institute of Technology, Rajapalayam
- On 18th September 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security** (CoE) at SRM University (Ramapuram Campus), Chennai
- On 20th November 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security** (CoE) at St.Joseph's Institute of Technology, Chennai

Our Company Supraja Technologies is one of the emerging startup in Andhra Pradesh with 4.8 Google Ratings



Link : https://bit.ly/SuprajaGoogle

AND

also check our Company CEO Instagram Profile for our recent more success stories:

https://www.instagram.com/chaluvadisantosh/



Santosh Chaluvadi

Founder & CEO

Supraja Technologies

He is a 28-year-old entrepreneur, one of the India's efficient Cyber Security Analyst and also he is an expert Digital Marketer as well. He is a digital marketer by profession and security enthusiast by passion. He primarily focuses on content building, testing and monetization of blogs. He has successfully developed many websites and done the security testing himself to ensure that the user's data is in safe hands and their privacy is protected. He is very active on social media and shares lot of tech stuff with his followers. The young student hacker has solved many issues with the vulnerabilities present in various websites and databases, given a solution in clearing the loopholes in order to protect the data to be leaked from the databases. Besides Ethical Hacking & Cyber Security, he also has a passion in Blogging & Digital Marketing.

While pursuing his engineering itself, he has trained many young generation people/students of more than 3500+ from various parts across Andhra Pradesh through his workshops, seminars, courses in Cyber Security and this makes him one of the youngest student trainers in India.

At the age of 20 he conducted his first workshop on Blogging & Ethical Hacking which was the beginning to his success in this field and right now he has handful of workshops to train students, government and corporate organizations as well in Andhra Pradesh

& Telangana. He is the only student trainer who started conducting workshop for his peers, professors and for corporates.

The year 2016 gave me the conviction and confidence to notch up whatever I was doing. I'd organized a 50-hour marathon event on Ethical Hacking and Cyber Security in PSCMR College, Vijayawada that went on to bag to achieve Limca Book of Records for non-stop longest duration workshop. The impact we were making was clearly visible by now. Diverse people from Jammu & Kashmir in the North to Kanyakumari in the South had attended the event. Some of the Government of India officials took notice of this record and invited our team for couple of conferences at New Delhi. The country's esteemed institutions were reaching out to me to conduct training, workshops, and events on a myriad of subjects related to online security. MultiNational Corporations (MNC's) had begun to consult me on the Cyber Security of their systems.

✞ Life changing solution/service :

After working on R&D for around 2 years, finally in the mid 2019 we have successfully developed a service/solution of various techniques and strategies for the Film Industry through which he can kill piracy of any film in online up to 35% right now. This betaservice is being appreciated & adopted by various Tollywood Film Industry Producers & Hero's to safeguard their film from piracy in online and to gain more profits.

By the end of 2033 our vision is to rollout a complete full packed service/solution where we can kill piracy entirely 100% everywhere in online for sure.

Appreciation:

Received a great appreciation from our 1st Tollywood Film Industry client Mr.Saptagiri for providing our Anti-Piracy betaservice for his film VAJRA KAVACHADARA GOVINDA

✞ Our Company Achievements:

- On 17th August 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security** (CoE) at Ramco Institute of Technology, Rajapalayam

- On 18th September 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security** (CoE) at SRM University (Ramapuram Campus), Chennai
- On 20th November 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security** (CoE) at St.Joseph's Institute of Technology, Chennai

✞ Records, Appreciations, Awards & Recognitions etc at a glance :

1. Holds a National Record in "Limca Book of Records – 2017"
2. Ex-Associate Member for National Cyber Safety and Security Standards (NCSSS)
3. Steering Committee Member for United Conference on Cyber Space (UNITEDCON 2020)
4. Judge for the Grand Finale of SIH (Smart India Hackathon 2024) Software Edition for the Nodal Centre at Sri Sai Ram Engineering College, Chennai which is an initiative by Ministry of Education (Govt. of India) and AICTE
5. Received Appreciation from Mr.Amit Narayan, Executive Director at Rajahmundry Asset of Oil and Natural Gas Corporation Limited (ONGC) on 16th December, 2022 for training their employees on Cyber Security
6. Awarded as a "Karmaveer Chakra - 2019", on 12th October 2019 at IIT Delhi, which was instituted by iCONGO in partnership with the United Nations
7. Received Appreciation from Mr.Sandeep Rathore, Commissioner of Police, Chennai on 2nd March 2024 for the exceptional commitment and invaluable contribution as a Member of JURY at Greater Chennai Police Cyber Hackathon 3.0
8. Evaluator for the "Innovative Bharat" which is organized by AICTE and Ministry of Education held on 6th January 2024
9. Judge for the Grand Finale of SIH (Smart India Hackathon 2023) Senior Software Edition for the Nodal Centre at PVPSIT, Vijayawada which is an initiative by Ministry of Education (Govt. of India) and AICTE
10. Appointed as a Member for Board of Studies on 19th April 2025 for the Departments of Cyber Security, Data Science and AIML at Bapatla Engineering College, Bapatla
11. Appointed as a Member for Board of Studies on 12th March 2025 for the Department of Cyber Security at St. Joseph's Institute of Technology, Chennai
12. Appointed as one of the Industry Academia Advisory Council Member on 30th September 2023 for the Department of Cyber Security at VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad
13. Appointed as a Member for Board of Studies on 4th May 2023 for the Department of Cyber Security at Madanapalle Institute of Technology & Sciences, Madanapalle
14. Appointed as a Member for Board of Studies for the Department of MCA at NRI Institute of Technology, Perecherla (Guntur)
15. Awarded as a "Social Media Influencer - 2019", on 30th June 2019 by Jignasa in association with Government of Andhra Pradesh

16. Nominated for “INDIA 500 CEO AWARD 2019”
17. Invited & Interviewed by ETV Andhra Pradesh news channel on 27th July, 2019 for a Special Story Interview on “Spy Apps”
18. Appreciated by Mr.Sridhar, Sub-Inspector of Police at Central Crime Branch, Vijayawada on 23rd October, 2018 for exclusively training him on Special Investigation Course, which will help him to solve the cyber crime cases easily
19. Received a great appreciation from our 1st Tollywood Film Industry client Mr.Saptagiri, for providing Anti-Piracy betaservice for his movie VAJRA KAVACHADARA GOVINDA

✞ Something we are proud of :

We are very happy to inform you all that our company, Supraja Technologies has been shortlisted for **"Top 50 Tech Companies" award 2019**, conferred at InterCon - Dubai, UAE.

Supraja Technologies is one out of thousands of startup companies that were initially screened by InterCon team of 45+ research analysts over a period of three months and the final shortlist includes 150+ firms and we are very proud to inform you all that our company Supraja Technologies also happens to be a part of the same.

Some Glimpses of our Journey



Mr.Santosh Chaluvadi – CEO, Supraja Technologies
Giving hands-on Cyber Security training workshop to the CSE students at IIT Kharagpur



Mr.Santosh Chaluvadi – CEO, Supraja Technologies was
Invited & Interviewed by ETV Andhra Pradesh news channel on 27th July, 2019 for a Special Story Interview on “Spy Apps”





Mr.Santosh Chaluvadi – CEO, Supraja Technologies was awarded as a **"Social Media Influencer 2019"** in recognition of his remarkable achievements in the social media as a part of First International Social Media Festival on 30th June 2019 by Jignasa in association with Government of Andhra Pradesh



On 23rd October 2018 Mr.Santosh Chaluvadi, CEO - Supraja Technologies and Mr.Krishna Chaitanya, CTO - Supraja Technologies had successfully completed delivering Special

Investigation Course training in Cyber Security to Mr.Sridhar Garu, Sub-Inspector of Police at Central Crime Branch, Vijayawada which will help him to solve the cyber crime cases easily



ETV Andhra Pradesh News Channel interviewed Mr.Santosh Chaluvadi, CEO - Supraja Technologies for his achievements in the domain of Cyber Security & Digital Marketing



Supraja Technologies was invited by Indian Air Force (Air Wing NCC) to deliver a session on Latest Cyber Crimes & Awareness for the NCC cadets, staff and officers on 4th July, 2019



Supraja Technologies – CEO, CTO & CMO with Indian Air Force (Air Wing NCC) Group Captain Sandeep Gupta.

We thank Mr.Sandeep Gupta for inviting us on 4th July, 2019 to deliver a session on Latest Cyber Crimes & Awareness for the Indian Air Force (Air Wing NCC) cadets, staff & officers

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

PROJECT REPORT ON

INSIDER THREAT PREDICTOR SYSTEM

Submitted by

Chittawar Anushri - AD23 - ST#IS#8429

Under the Supervision of

UPENDRA
Senior Security Analyst

KRISHNA
Security Analyst

Registered And Head OfficeD.NO:
11-9-18, 1st Floor,

Majjivari Street, Kothapeta,
Vijayawada - 520001.

+91 9550055338 / +91 7901336873

contact@suprajatechnologies.com

Table of Contents

1. Abstract.....1

2. Introduction.....1

2.1 Overall Description ----------1

3. Existing System.....2

4. Proposed System 3

5. System Design.....3

5.1 Feasibility Study 3

5.1.1 Technical Feasibility ----------3

5.1.2 Operational Feasibility ----------3

5.1.3 Economic Feasibility ----------3

5.1.4 Legal Feasibility ----------4

5.2 Input and Output Design.....4

5.2.1 Input Design ----------4

5.2.2 Objectives ----------4

5.2.3 Output Design ----------4

6. Implementation 5

6.1 Module Description ----------5

6.2 System Architecture ----------5

7. Algorithm Implementation 7

7.1 Login Anomaly Detection Flow----------7

7.2 Suspicious Process Detection Flow ---- 9

7.3 USB Activity Risk Analysis ---- 11

7.4 Risk Scoring & Threat Classification Logic ---- 11

8. System Design Diagrams **12**

8.1 Data Flow Diagram ---- 12

8.2 Use Case Diagram ---- 12

8.3 Sequence Diagram ---- 13

8.4 Activity Diagram ---- 14

8.5 Class Diagram ---- 15

8.6 Component Diagram ---- 16

8.7 ER Diagram ---- 17

9. Requirement Specification **17**

9.1 Functional Requirements ---- 17

9.2 Software Requirements ---- 17

9.3 Operating Systems Supported ---- 18

9.4 Technologies and Languages Used ---- 18

9.5 Hardware Requirements ---- 18

10. System Testing **18**

10.1 Types of Test ---- 18

10.2 Test Strategy and Approach ---- 19

10.3 Test Results ---- 20

10.4 Execution ---- 20

11. Conclusion **26**

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

1. Abstract

Insider threats pose a significant risk to organizational security due to the trusted access insiders possess to critical systems and data. Traditional perimeter-based security mechanisms are often ineffective in detecting malicious or negligent insider activities. This project presents the design and implementation of a **Windows-based Insider Threat Predictor System** that monitors user behavior through real-time analysis of Windows Security Event Logs.

The system focuses on detecting high-risk insider activities such as repeated failed login attempts, administrative privilege escalation, and removable media (USB/pendrive) insertion and access. A rule-based risk scoring mechanism is used to quantify user behavior by assigning weighted risk values to security-relevant events. These risk scores are continuously updated and visualized through an interactive graphical dashboard developed using Python, Tkinter, and ttkbootstrap.

By providing real-time visibility into insider activities and highlighting high-risk behavior, the system demonstrates a foundational User and Entity Behavior Analytics (UEBA) approach without relying on machine learning techniques. Although the proposed solution does not perform predictive analytics, it effectively illustrates how insider threat detection can be achieved using behavioral monitoring and risk-based analysis. The system can serve as a baseline for future enhancements involving anomaly detection, behavioral baselining, and machine learning–driven insider threat prediction.

2. Introduction

With the increasing dependence on digital systems and sensitive data, organizations face growing security threats not only from external attackers but also from individuals operating within the organization. These individuals, commonly referred to as *insiders*, already possess legitimate access to systems, applications, and data, making insider threats particularly dangerous and difficult to detect. Insider threats may arise from malicious intent, negligence, or compromised credentials, and they often result in severe data breaches, financial losses, and reputational damage.

Traditional security solutions such as firewalls, intrusion detection systems, and antivirus software are primarily designed to protect against external threats. While effective at network boundaries, these mechanisms are often inadequate for identifying suspicious behavior performed by authorized users. As a result, there is a growing need for security systems that focus on monitoring user behavior and identifying anomalies or high-risk activities from within the system.

This project presents the development of an **Insider Threat Predictor System** for Windows environments that monitors user activities through Windows Security Event Logs. The system analyzes critical security events such as login attempts, administrative privilege escalation, and removable media (USB/pendrive) usage. Based on these events, a rule-based risk scoring mechanism is applied to evaluate the potential threat level associated with a user's actions.

2.1 Overall Description

The Insider Threat Predictor System is a Windows-based security monitoring tool designed to detect suspicious insider activities in real time. It analyzes Windows Security Event Logs to identify behaviors such as failed logins, administrative privilege escalation, and USB (pendrive) usage. A rule-based risk scoring mechanism inspired by AI-driven UEBA concepts is used to model user behavior and assess threat levels. The system provides an interactive

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

dashboard for visualizing activities, risks, and alerts to support security decision-making.

3. Existing System

In most organizations, insider threat detection primarily relies on traditional security mechanisms such as firewalls, antivirus software, and basic log monitoring tools. These systems are mainly designed to detect external attacks and known malware signatures rather than malicious or negligent actions performed by authorized users. Log analysis is often manual, fragmented, or reactive, making it difficult to identify suspicious insider behavior in real time. Additionally, existing systems usually lack behavioral correlation, risk scoring, and effective visualization, which results in delayed detection of insider threats.

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

4. Proposed System

The proposed Insider Threat Predictor System introduces a behavior-based monitoring approach for detecting insider threats in Windows environments. It continuously analyzes Windows Security Event Logs to monitor user activities such as login attempts, privilege escalation, and removable media usage. A rule-based risk scoring mechanism, inspired by AI-driven UEBA concepts, is applied to evaluate the threat level of user behavior. The system provides a real-time graphical dashboard with alerts and risk indicators, enabling faster identification and response to potentially malicious insider activities.

5. System Design

The system design describes the overall structure and functioning of the Insider Threat Predictor System. It defines how the system components interact, how data flows through the system, and how inputs are processed to produce meaningful outputs. The design focuses on real-time monitoring, risk-based analysis, and effective visualization of insider activities in a Windows environment.

5.1 Feasibility Study

A feasibility study was conducted to evaluate whether the proposed Insider Threat Predictor System is technically, operationally, economically, and legally viable before development and deployment.

5.1.1 Technical Feasibility

The proposed system is technically feasible as it is developed using widely supported and proven technologies. The system is implemented in Python and uses standard libraries such as win32evtlog for accessing Windows Security Event Logs and Tkinter with ttkbootstrap for graphical user interface development. Since the solution operates on Windows platforms and leverages built-in system logs, no additional hardware or specialized infrastructure is required. The system can run on standard computers with administrative privileges, making technical implementation practical and achievable.

5.1.2 Operational Feasibility

Operational feasibility is high because the system is designed to be user-friendly and easy to operate. The graphical dashboard provides clear controls such as start and stop monitoring options, real-time activity logs, and alert notifications. Security administrators or system operators can easily understand and use the system without requiring extensive technical training. The automated monitoring and alert generation reduce manual effort, improving operational efficiency in detecting insider threats.

5.1.3 Economic Feasibility

The system is economically feasible as it uses open-source technologies and does not require any paid software licenses. Python, Tkinter, and Windows event logging tools are freely available. Since the system runs on existing hardware and operating systems, there are no additional costs related to infrastructure or maintenance. This makes the solution cost-effective and suitable for small to medium organizations as well as academic environments.

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

5.1.4 Legal Feasibility

The proposed system is legally feasible as it operates within the boundaries of organizational security monitoring policies. It analyzes system-generated security logs and does not intrude into personal data such as private communications or personal files. USB monitoring and login tracking rely on standard Windows auditing mechanisms, ensuring compliance with organizational IT policies. However, proper authorization and adherence to data privacy regulations are required before deployment in real-world environments.

5.2 Input and Ouput Design

Input and output design focuses on how data enters the system, how it is processed, and how results are presented to users in a clear and meaningful manner.

5.2.1 Input Design

The primary input to the system is data collected from the Windows Security Event Log. This includes system-generated events related to user authentication, privilege escalation, and removable media usage. These events are automatically captured without manual user input. The system filters relevant event IDs and extracts essential information such as timestamp, event type, and severity level for further analysis.

5.2.2 Objectives

The objectives of the input design are:

1. To capture only security-relevant events from the Windows Security logs
2. To ensure accurate and real-time data collection
3. To minimize unnecessary or redundant log data
4. To provide structured input for effective risk analysis
5. This approach ensures that the system processes meaningful data while maintaining efficiency.

5.2.3 Output Design

The output of the system is presented through an interactive graphical dashboard. The dashboard displays real-time activity logs, user risk scores, and alert notifications. Outputs include tables showing detected events, alerts for suspicious activities, and summarized risk indicators for users. The output design emphasizes clarity, readability, and quick decision-making by using structured tables and visual indicators. This allows security personnel to easily identify high-risk behavior and respond promptly.

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

6. Implementation

6.1 Module Description

The Insider Threat Predictor System is implemented as a modular security monitoring platform where each component performs a distinct function while integrating seamlessly with others.

Modules:

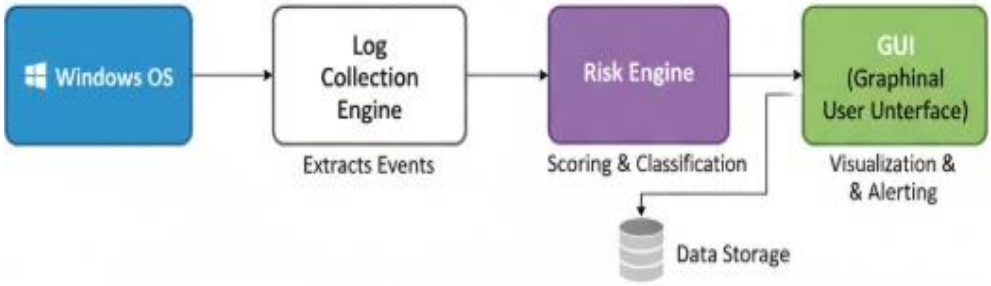
1. **GUI Module** – Provides an interactive dashboard using Tkinter and ttkbootstrap for real-time visualization of user activities, risk scores, and alerts.
2. **Log Collection Engine** – Continuously monitors Windows Security Event Logs to capture login attempts, privilege escalation events, and USB (pendrive) insertions.
3. **Risk Scoring Engine** – Applies rule-based and behavior-driven logic inspired by UEBA and AI concepts to calculate cumulative risk scores for users.
4. **Event Processor** – Filters, normalizes, and categorizes raw log events into meaningful security actions.
5. **Alert & Incident Manager** – Generates alerts when predefined thresholds are exceeded and logs incidents for investigation.
6. **Report Generator** – Produces summarized security reports containing detected events, risk levels, and incidents.

6.2 System Architecture

The system follows a three-layer architecture consisting of the data collection layer, processing layer, and presentation layer.

Figure 6.1: System Architecture

1. The Log Collection Engine extracts events from the Windows OS.
2. Events are sent to the Risk Engine for scoring and classification.
3. Processed data is stored and forwarded to the GUI for visualization and alerting.



Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

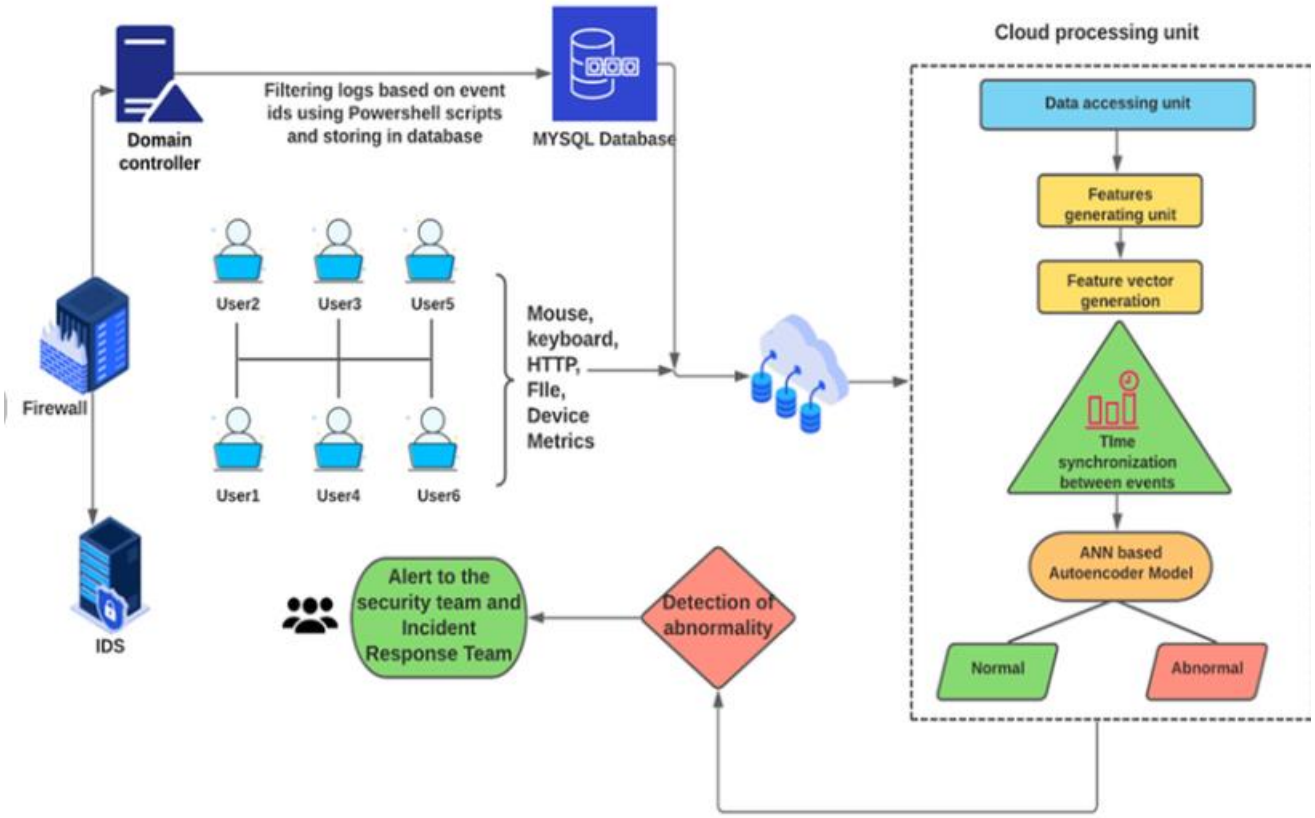


Figure 6.1: System Architecture

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

7. Algorithm Implementation

7.1 Login Anamoly Detection Flow

Login Anomaly Detection is used to identify abnormal or suspicious login behavior by analyzing user authentication events. Since insiders are already trusted users, abnormal login patterns such as unusual timing, repeated failures, or access from unexpected contexts are strong indicators of potential insider threats.

The system continuously monitors login activities collected from the operating system and authentication logs. Each login event is evaluated using predefined rules and assigned a risk score based on its behavior.

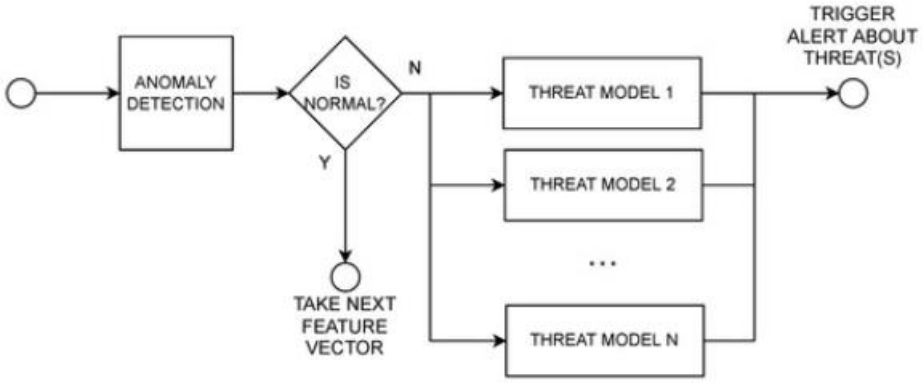
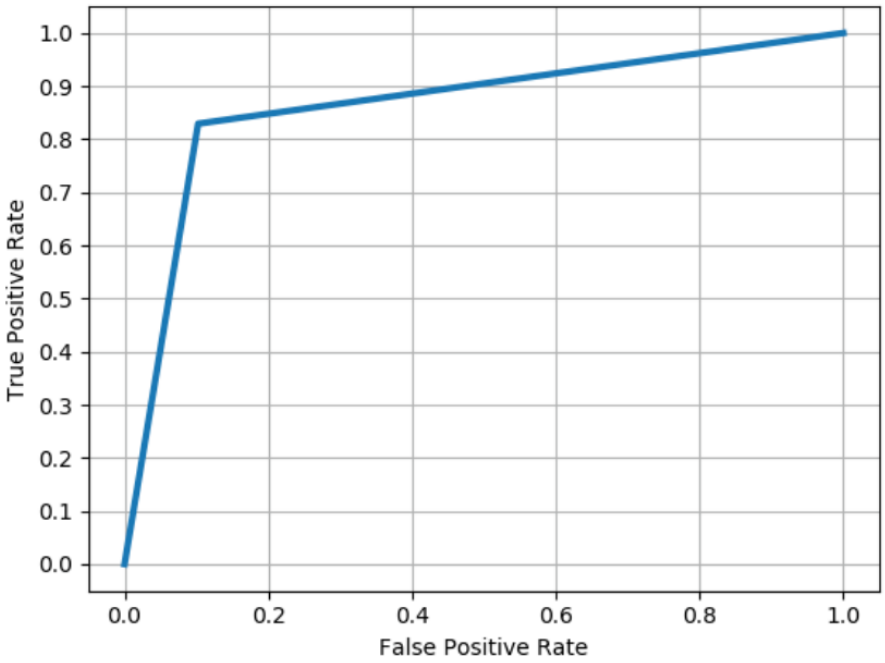


Figure 3. The model cascade used for anomaly detection and threat identification (there are two steps of classification in the proposed approach: anomaly detection and threat identification).

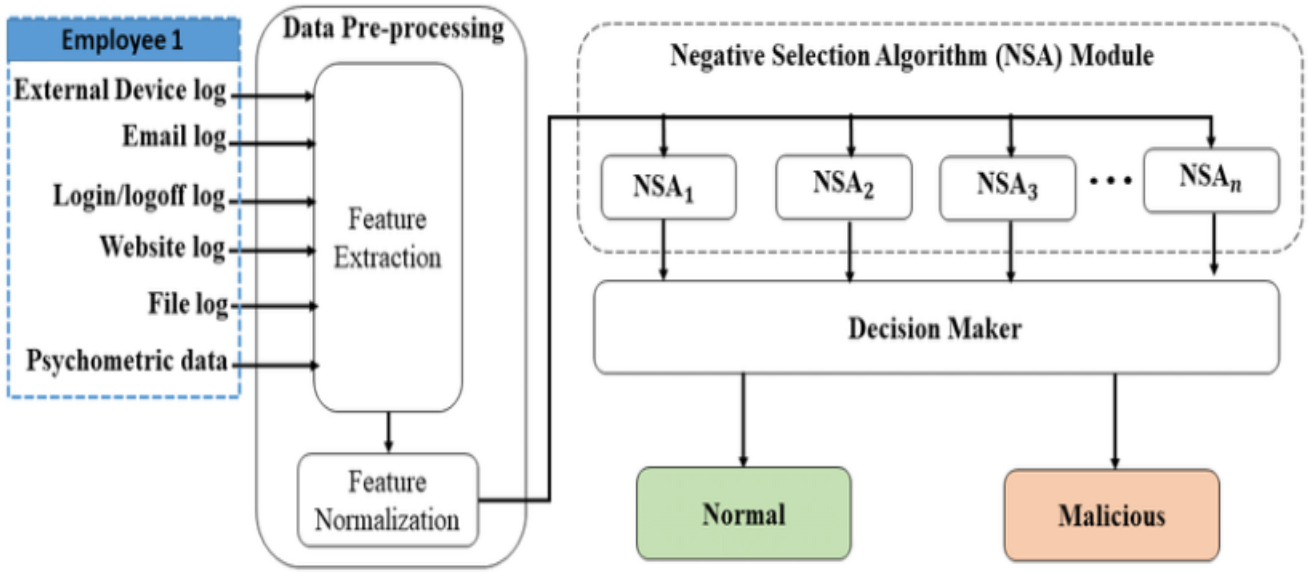
Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144



The ROC Curve

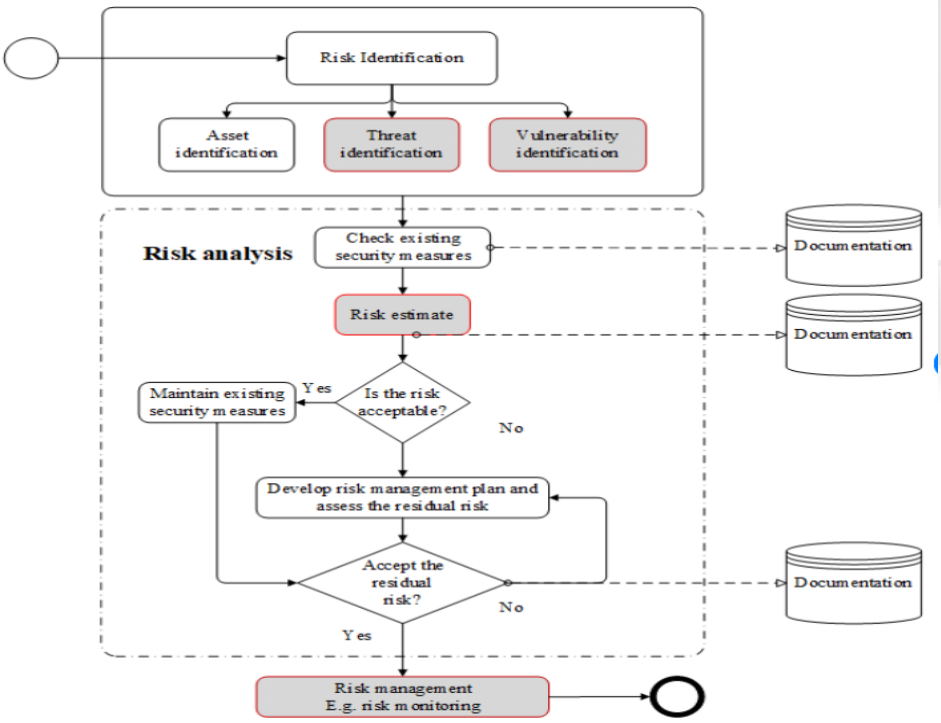
Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

7.2 Suspicious Process Detection Flow

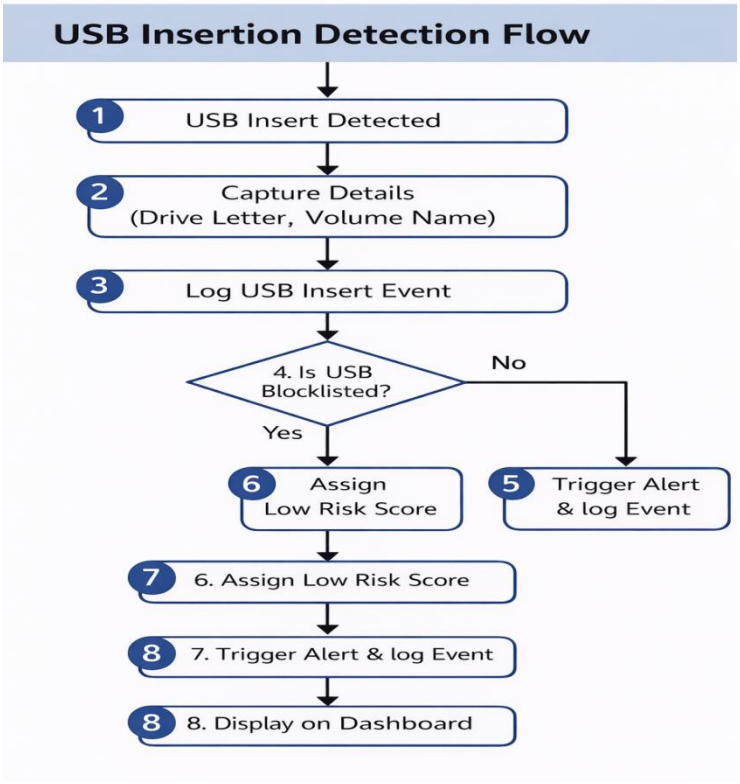


The image part with relationship ID rId52 was not found in the file.

7.3 USB Risk Analysis



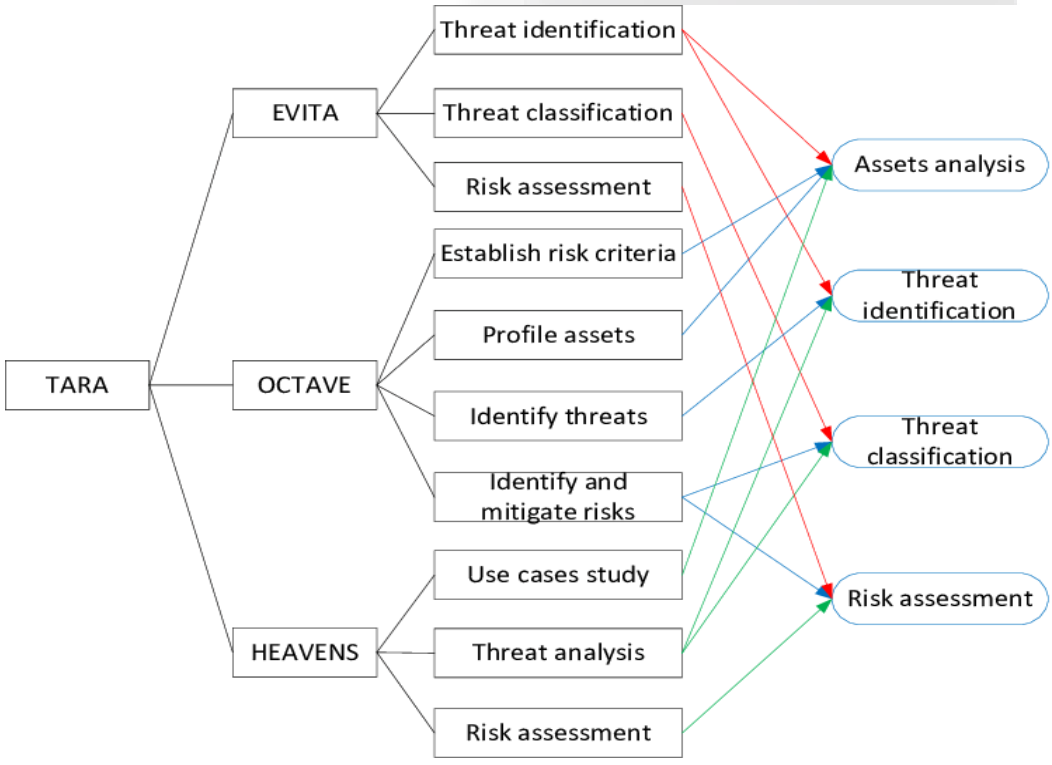
Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144



Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

7.4 Risk Scoring and Threat Classification Logic





8. System Design Diagrams

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

8.1 Data Flow Diagram

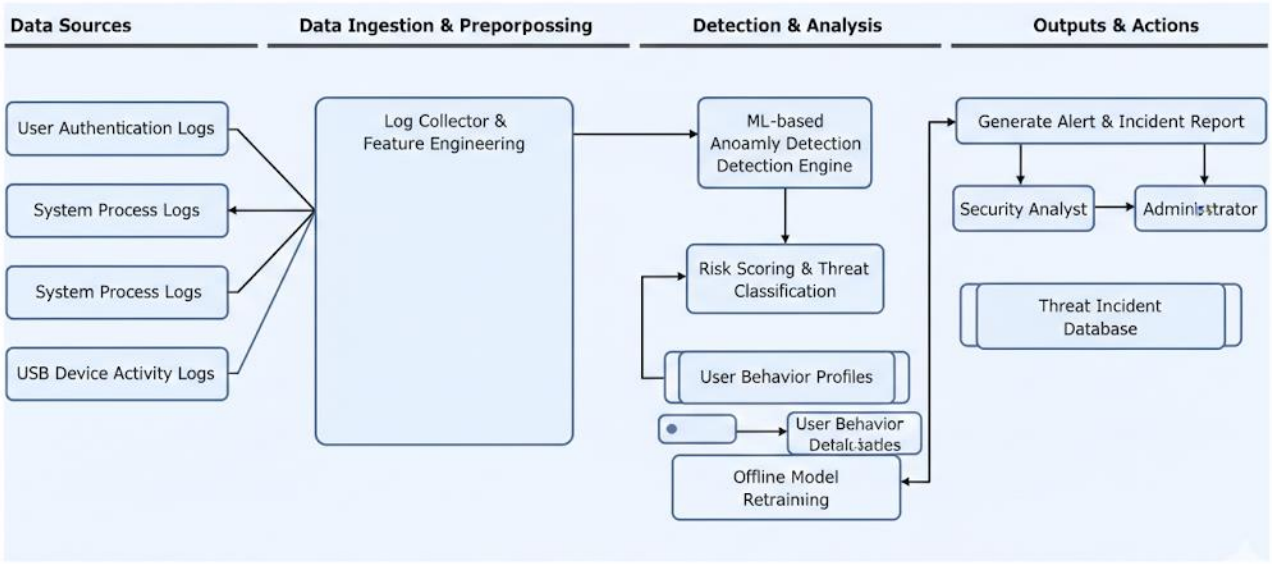


Figure 8.1: Data Flow Diagram (Level 0)

8.2 Use Case Diagram

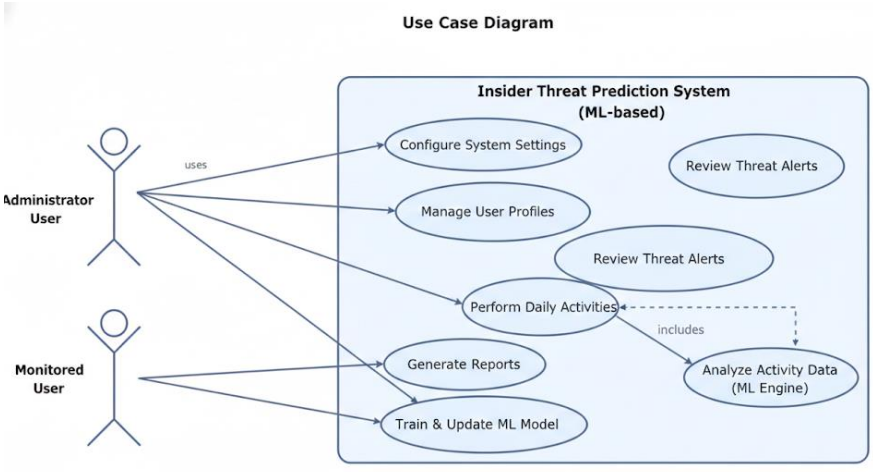


Figure 8.2: Use Case Diagram

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

8.3 Sequence Diagram

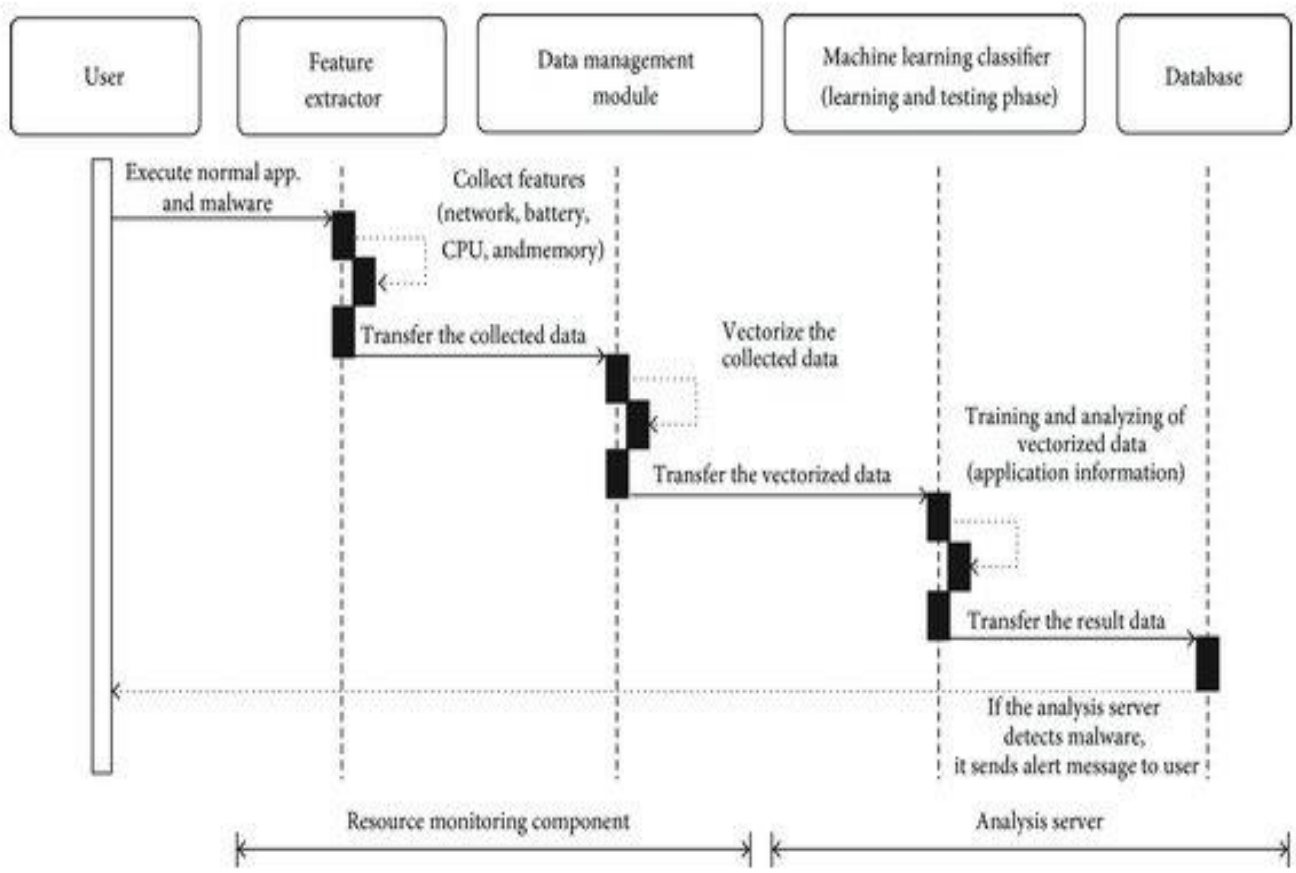


Figure 8.3: Sequence Diagram

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

8.4 Activity Diagram

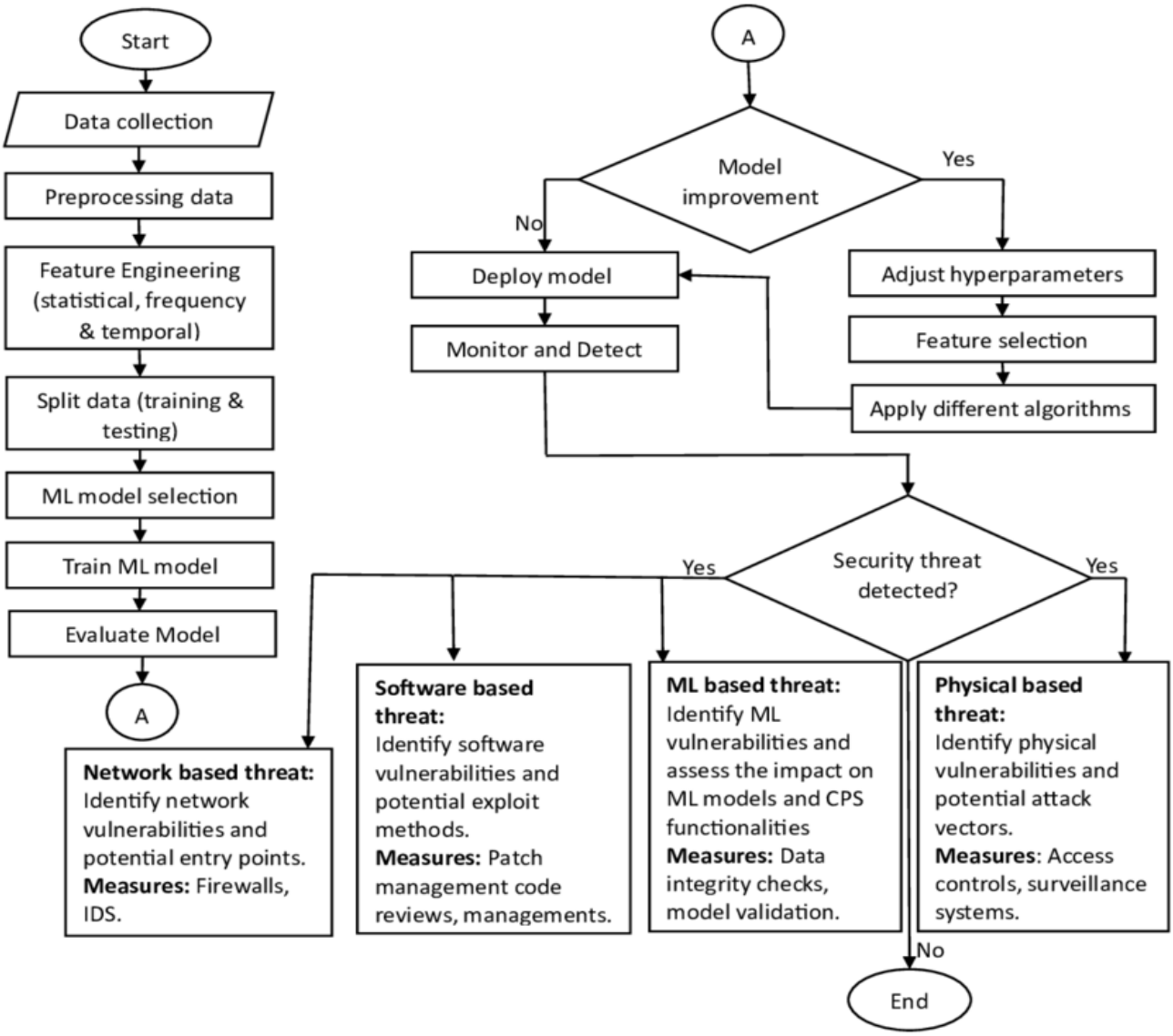


Figure 8.4: Activity Diagram

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

8.5 Class Diagram

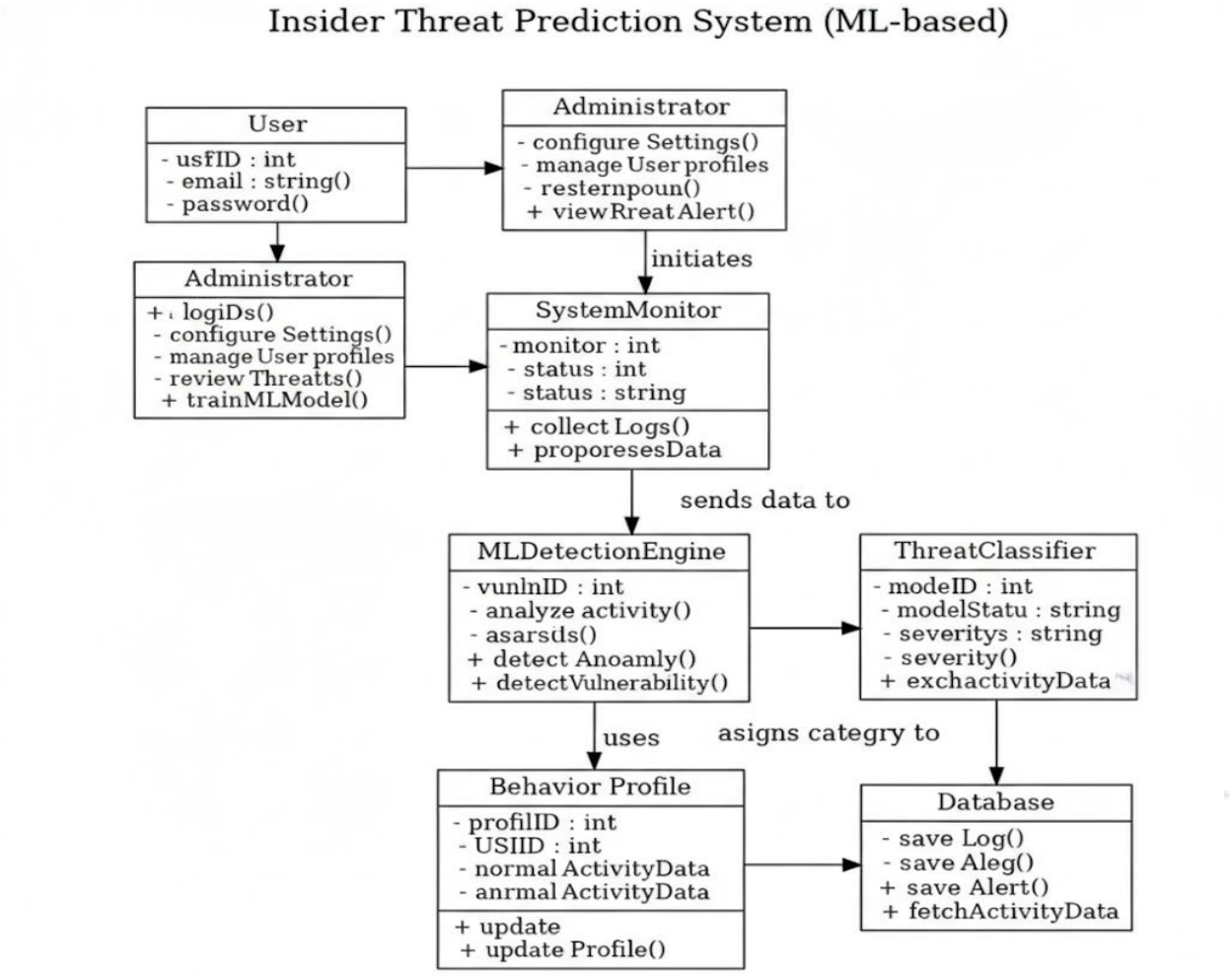
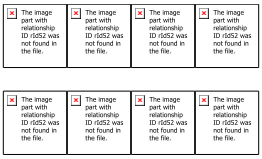


Figure 8.5: Class Diagram

8.6 Component Diagram



Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

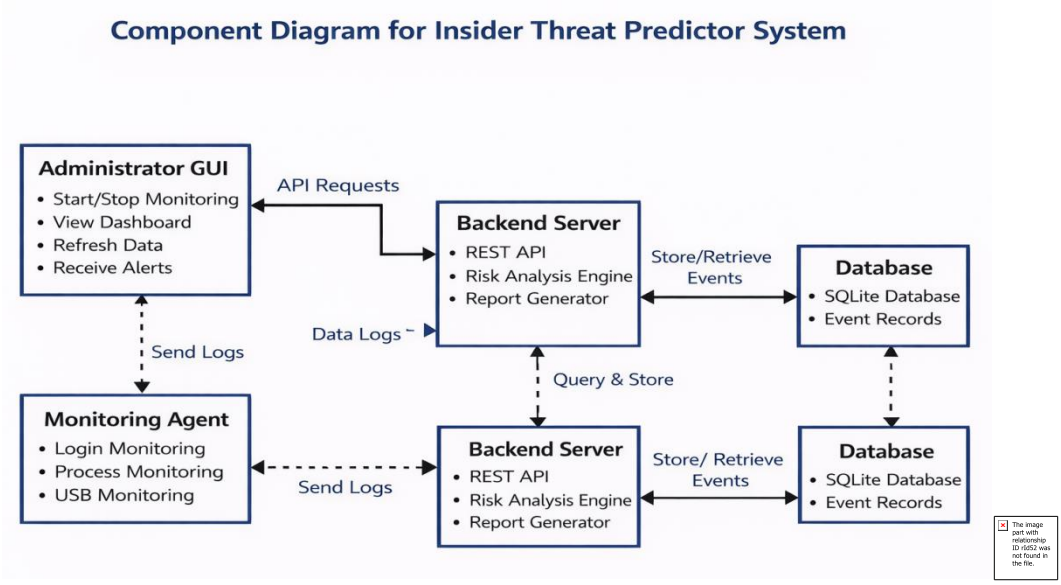


Figure 8.6: Component Diagram

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

8.7 ER Diagram

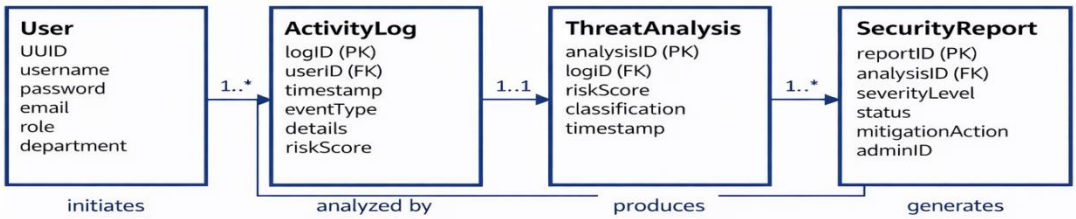


Figure 8.7: ER Diagram

9. Requirement Specification

9.1 Functional Requirements

The functional requirements define what the Insider Threat Predictor System must do in order to monitor, analyze, and predict insider threats within an organization.

- Monitor user login activities in real time
- Detect login anomalies such as repeated failures or abnormal access times
- Monitor suspicious processes and unauthorized application execution
- Detect USB insertion and removable media usage
- Capture system and user activity logs automatically
- Assign risk scores based on predefined behavioral rules
- Store all monitored events securely in the database

9.2 Software Requirements

Programming Language: Python 3.10 or higher
- Backend Framework: FastAPI
- Database: SQLite
- Frontend/Dashboard: Python (Tkinter / Custom GUI)
- Operating System APIs for log monitoring
- Libraries: OS, psutil, requests, sqlite3, datetime

x.

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

9.3 Operating Systems Supported

Windows 10 / Windows 11
- Linux (Ubuntu 20.04 and above)
- macOS (with Python support – limited monitoring features)

9.4 Technologies and Languages Used

Python
- REST API (HTTP/HTTPS)
- SQLite Database
- Rule-based Risk Scoring Logic
- System Log Monitoring
- File and Process Monitoring

9.5 Hardware Requirements

Processor: Dual-core CPU or higher
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: Minimum 10 GB free disk space
- Network: LAN / Internet connection for dashboard access

10. System Testing

10.1 Types of Test

During the development of the Insider Threat Predictor System, multiple levels of testing were conducted to ensure reliability, accuracy, and system stability. Testing was performed incrementally, from individual modules to the complete integrated system.

1. Acceptance Testing (Highest Priority)

Acceptance testing verified whether the system met the original objectives defined during project planning. The key acceptance criteria included real-time monitoring, risk scoring, event logging, and dashboard visualization.

Result:
The system successfully monitored insider activities, generated risk scores, and displayed alerts on the dashboard, meeting all acceptance requirements.

2. Security Testing *(Core Functionality)*

Security testing focused on detecting insider threat behaviors rather than external attacks. The system was tested against various internal misuse scenarios, including:

- Repeated failed login attempts
- Login outside normal working hours
- Execution of suspicious processes

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

- Unauthorized USB insertion
- Abnormal user behavior patterns

Result:
The system correctly identified suspicious activities and assigned appropriate risk levels.

3. System Testing *(High Priority)*

System testing validated the complete workflow, starting from event detection to database storage and dashboard visualization.

Result:
All system components worked together seamlessly, with accurate data flow and real-time updates.

4. Integration Testing *(Medium Priority)*

Integration testing ensured smooth communication between system components such as the monitoring agent, backend server, risk engine, database, and dashboard.

Result:
No major integration issues were observed during testing.

5. Unit Testing *(Medium Priority)*

Individual modules were tested independently to ensure correct functionality. Examples include:

- Login monitoring module
- USB monitoring module
- Process monitoring module
- Risk calculation logic

Result:
Each unit functioned correctly and produced expected results.

6. GUI Testing *(Medium Priority)*

The dashboard interface was tested for usability and responsiveness. Buttons such as Start Monitoring, Stop Monitoring, Refresh, and Exit were verified.

Result:
The dashboard was user-friendly and responded correctly to all user actions.

7. Performance Testing *(Low Priority)*

Performance testing evaluated system behavior under continuous monitoring and multiple event generation.

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

Result:
The system remained stable and responsive, suitable for academic and small-scale organizational use.

8. Regression Testing *(Supportive)*

Regression testing ensured that newly added features did not disrupt existing functionality.

Result:
Previously implemented features continued to work correctly after updates.

10.2 Test Strategy and Approach

Testing was conducted using simulated insider activities in a controlled environment. Various behavioral scenarios were generated to validate risk detection accuracy and minimize false positives.

10.3 Test Results

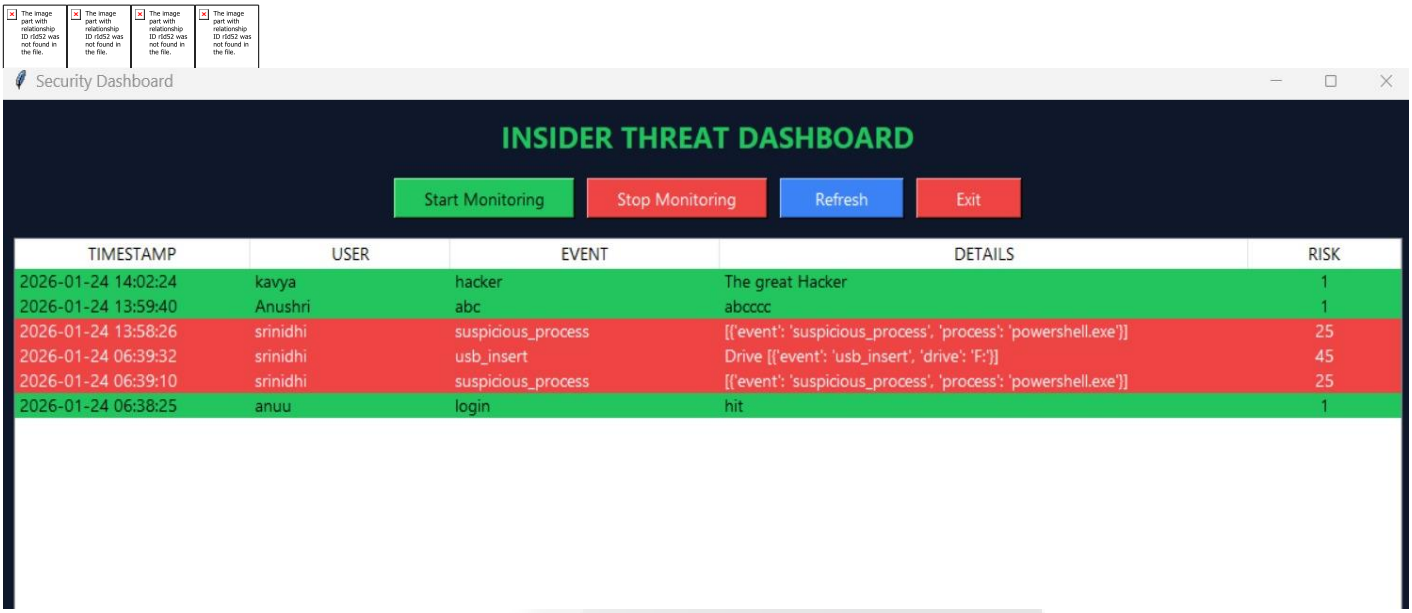
The system successfully met all functional and non-functional requirements and demonstrated reliable insider threat detection capability.

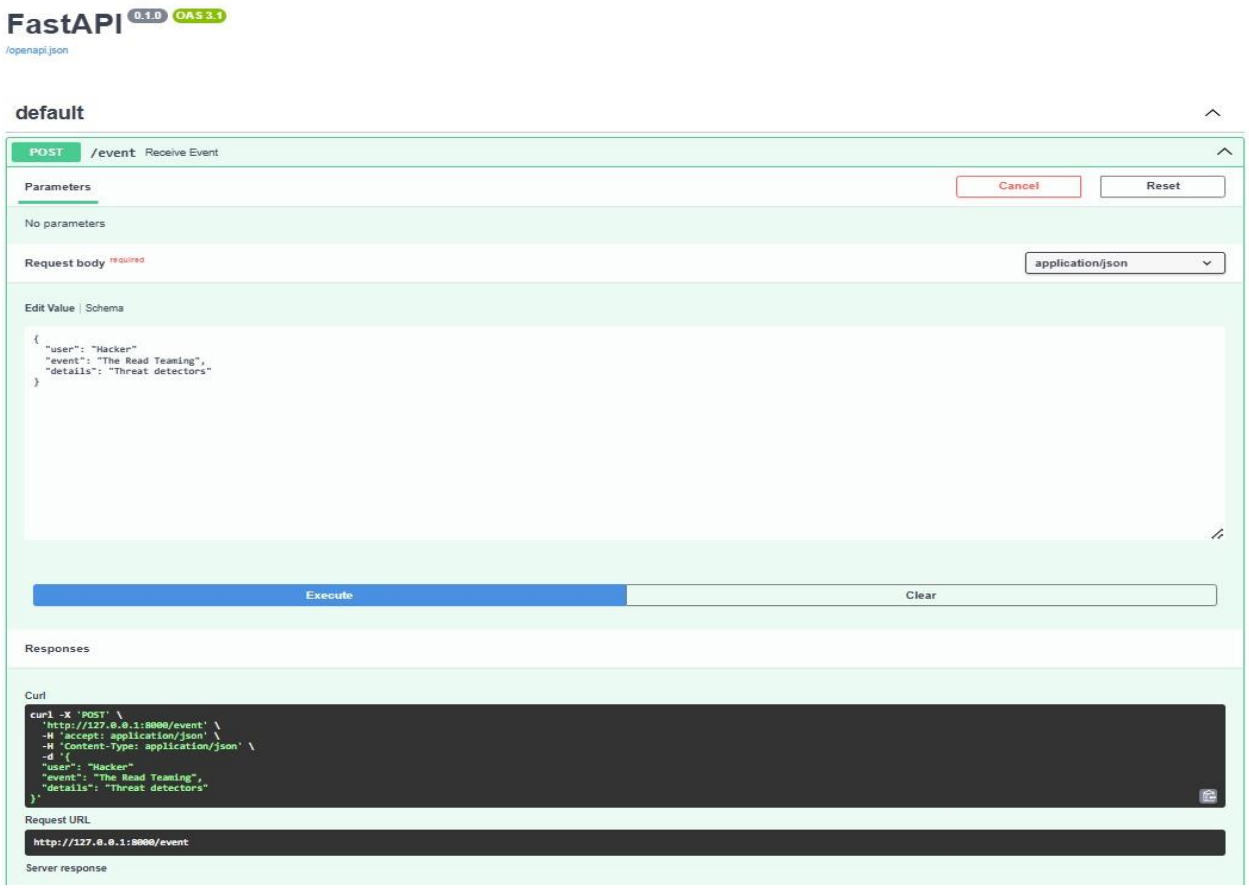
10.4 Execution

The Insider Threat Predictor System was executed on a Windows-based environment. Monitoring agents captured user activity in real time, analyzed events using the risk engine, and displayed results on the dashboard without errors.

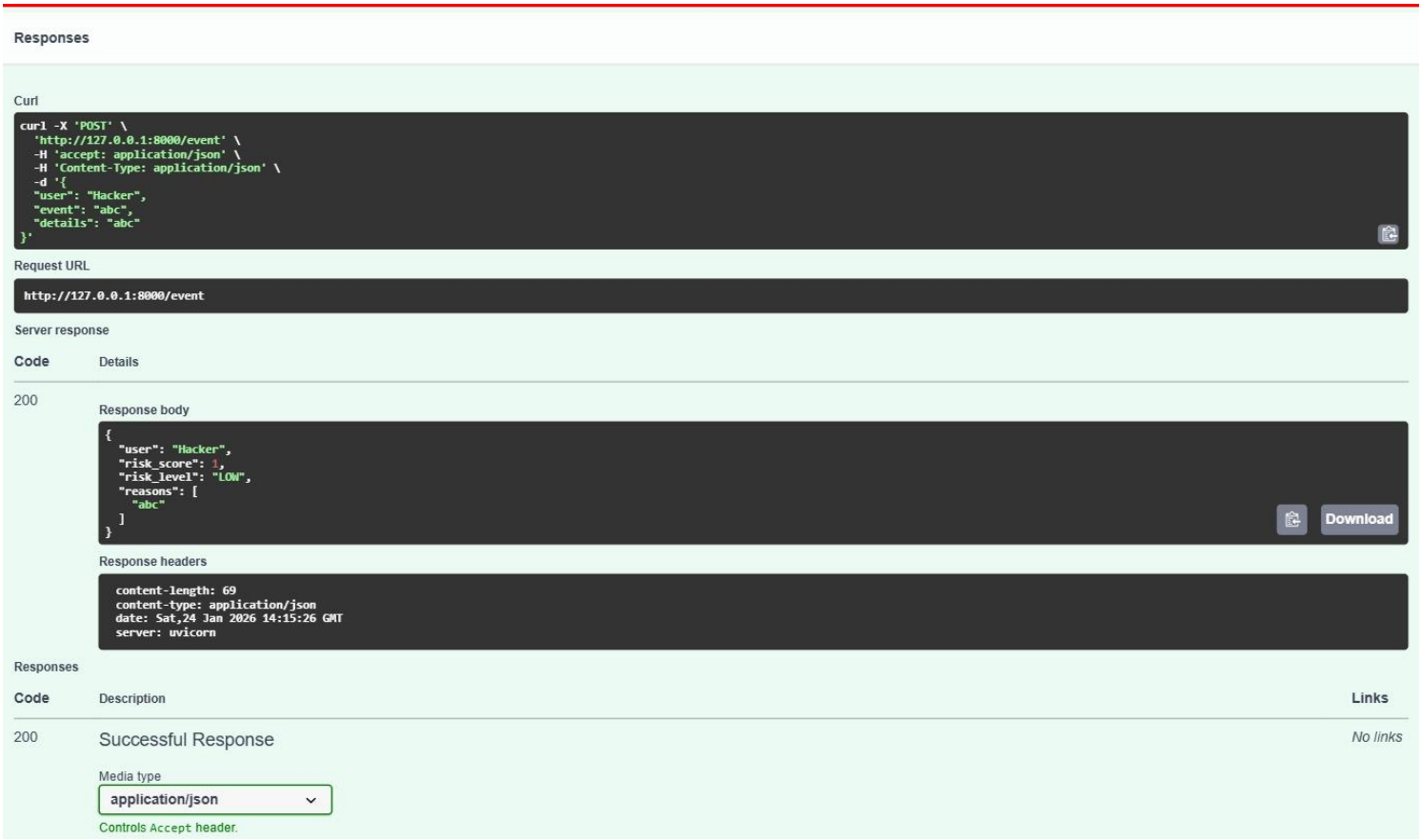
Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

GUI :





Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

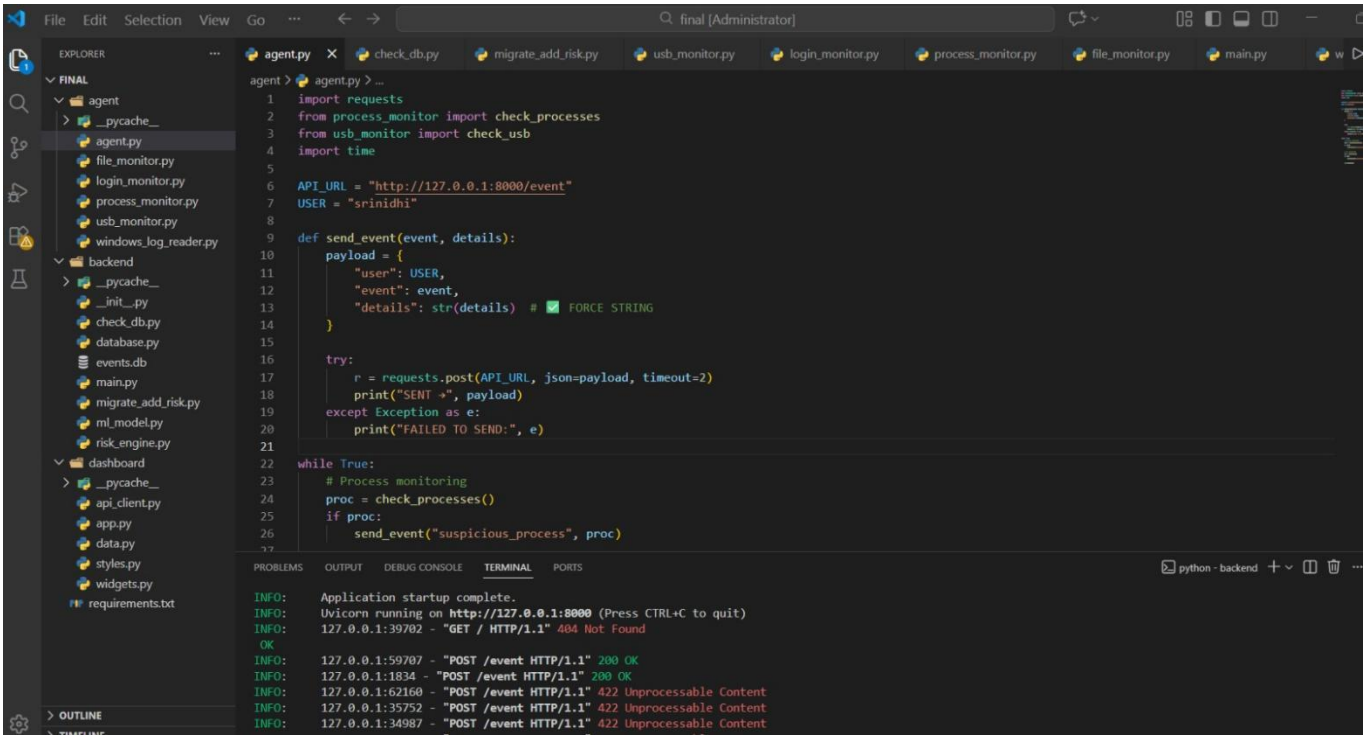



```
index=network sourcetype=network_traffic
| eval USB_Connected_Time = [search index=windows sourcetype=WinEventLog EventCode=20001
  | stats max(_time) as USB_Connected_Time
  | fields USB_Connected_Time]
| eval TimeSinceUSB = _time - USB_Connected_Time
| where TimeSinceUSB <= 300  // Time window in seconds (5 minutes)
| stats sum(bytes) as TotalBytes by src_ip, dest_ip, _time
| sort -TotalBytes
| table _time, src_ip, dest_ip, TotalBytes
```

Detecting USB based Threat

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

Proof of Concept:

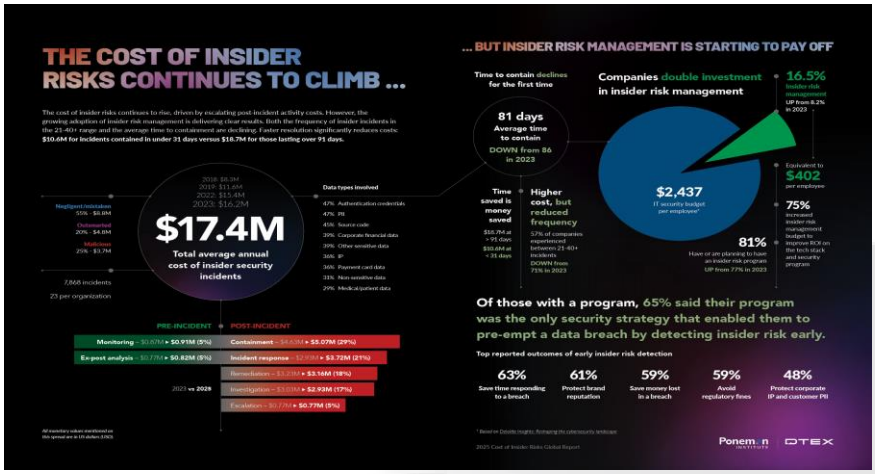


Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

Research Work:

Real-World Problems Addressed by Insider Threat Detection Systems

Insider threats are among the most challenging security risks faced by modern organizations because they originate from trusted users who already have legitimate access to systems and data. Unlike external attackers, insiders can bypass traditional perimeter defenses, making detection more complex.



1. Data Theft Using Removable Media

Employees may intentionally or unintentionally copy sensitive data using USB drives. Such incidents can lead to data leakage, intellectual property theft, or regulatory violations.

How this system helps:
The system detects USB insertion events in real time, logs activity, assigns risk scores, and alerts administrators when suspicious removable media usage is detected.

2. Misuse of Legitimate User Credentials

Compromised or misused credentials allow insiders to access systems outside normal working hours or repeatedly attempt logins, which may indicate malicious intent.

How this system helps:
Login anomaly detection identifies unusual login behavior, failed login patterns, and abnormal access times, helping administrators detect compromised accounts early.

3. Execution of Unauthorized or Malicious Processes

Insiders may execute unauthorized software, scripts, or administrative tools that can compromise system integrity.

Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144

How this system helps:
Process monitoring detects suspicious or blacklisted applications and assigns higher risk scores based on behavior patterns.

4. Lack of Visibility into Internal Activities

Many organizations lack centralized monitoring of internal user activities, resulting in delayed detection of insider threats.

How this system helps:
The centralized dashboard provides real-time visibility of all monitored events, enabling proactive security monitoring.

5. Delayed Incident Response

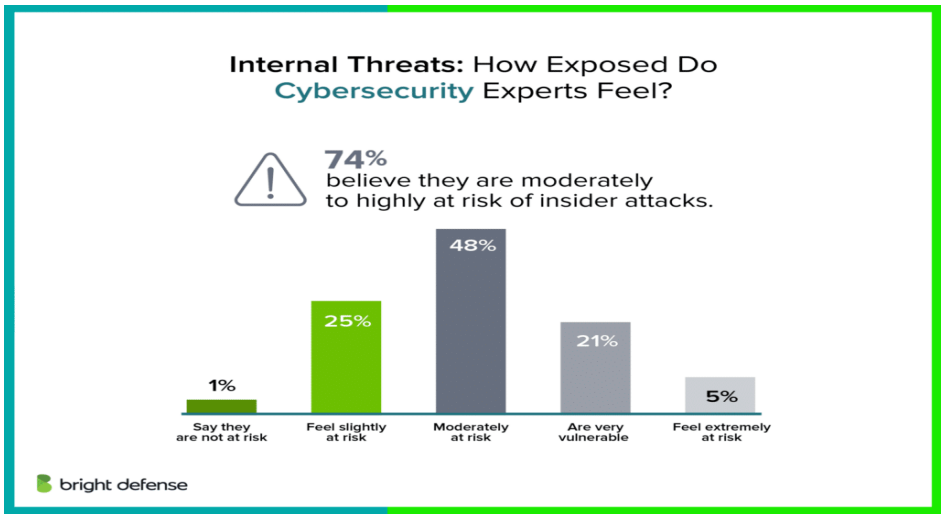
Without automated detection, security teams often respond only after significant damage has occurred.

How this system helps:
The risk-based alerting mechanism allows administrators to identify high-risk users and respond before incidents escalate.

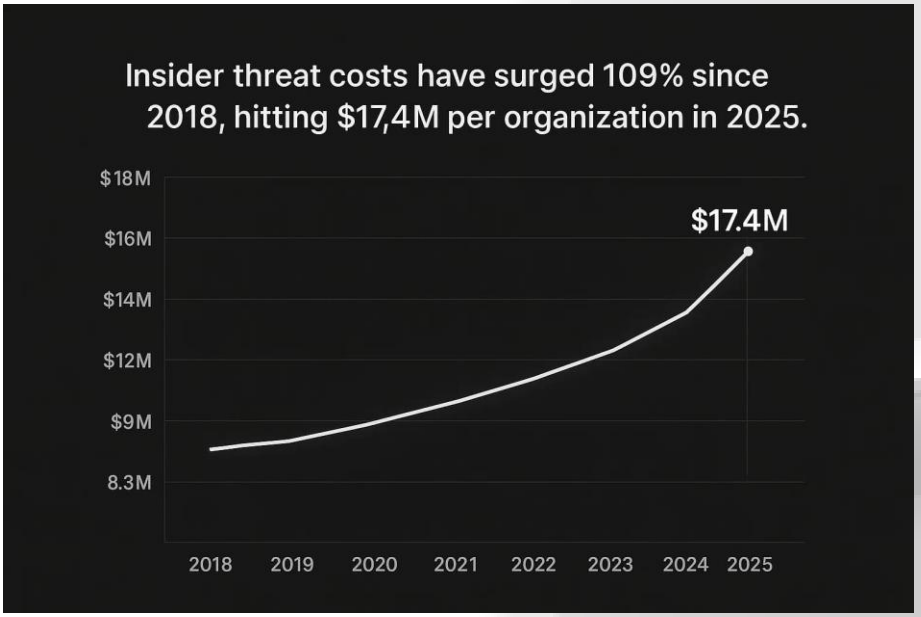
6. Compliance and Audit Challenges

Organizations must comply with regulations requiring monitoring and auditing of user activities (e.g., data protection policies).

How this system helps:
The system maintains detailed event logs and reports that can be used for auditing and compliance verification.



Company Identification Number (CIN Number) : U72900AP2019PTC111909
PAN Number : AAICC1770E, TAN Number : HYDC10078F
ISO Certificate Number : 031122019107
MSME Registration Certificate No : UDYAM-AP-06-0035821
Supraja Technologies Trademark Application ID No for Class 41 : 4146139
Supraja Technologies Trademark Application ID No for Class 42 : 4146144



Conclusion

The Insider Threat Predictor System successfully demonstrates an effective approach to identifying and analyzing security risks originating from within an organization. Unlike traditional security mechanisms that focus primarily on external attacks, this system emphasizes continuous monitoring of trusted users by observing login behavior, process execution, USB activity, and system events.

By integrating real-time monitoring agents with a centralized backend and risk analysis engine, the system is capable of detecting anomalous user behavior and assigning appropriate risk scores. The use of a structured database ensures secure storage of events, while the dashboard provides administrators with clear visibility into potential insider threats through real-time alerts and risk-based visualization.

The project achieves its core objectives of early threat detection, activity tracking, and centralized monitoring using a modular and scalable architecture. Although the current implementation relies on rule-based risk scoring, the design allows seamless integration of machine learning models for advanced behavioral prediction in future enhancements.

Overall, the Insider Threat Predictor System provides a reliable, efficient, and extensible solution for mitigating insider security risks and serves as a strong foundation for further research and real-world deployment in enterprise environments.